

Caleb Adams

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LinkedIn: [Caleb Adams](#) | Portfolio: [Engineering Portfolio](#) | Website: [Caleb Adams](#)

PROFESSIONAL SUMMARY

Mechanical engineering student with hands-on experience in HVAC installation, CAD design, prototyping, machining, and embedded systems. Passionate about product design, manufacturing, and solving real world engineering problems through iterative design and testing.

EDUCATION

The Ohio State University: Columbus, Ohio

Bachelor of Science in Mechanical Engineering, Business Minor, Expected May 2028

Relevant Courses: Linear Algebra and Differential Equations, Thermodynamics, Dynamics, Mechanics of Materials, Statics, Numerical Methods, Calculus 1-3, Physics I, II, Material Science.

EXPERIENCE

Adams Heating and Cooling

Cincinnati, OH

HVAC Intern

May 2025-August 2025, May 2026-August 2026

- Installed and serviced residential and commercial HVAC systems, including AC units, heat pumps, and mini splits.
- Fabricated refrigerant piping using pressing and bending equipment to minimize oxidation and eliminate debris.
- Performed nitrogen pressure testing, vacuum evacuation, and refrigerant charging.
- Completed electrical connections from disconnect boxes to unit, including 460 V three phase systems.
- Operated HVAC installation equipment, sheet metal fabrication tools, power tools safely and efficiently.

PROJECTS

Machine Piston

January 2026-March 2026

- Manufactured a working aluminum-brass piston assembly using manual lathes and milling machines maintaining a tolerance of ± 0.001 in.
- Used techniques such as facing, reaming, and threading to shape the piston block, connecting rod, and head.
- Interpreted engineering drawings and used dimension analysis throughout the machining process.
- Assembled and functionally tested the completed piston assembly, achieving smooth operation at 320 RPM.

Project Sleep: Smart Sleeping Device

January 2025-May 2025

- Designed CAD models and developed a sleep enhancing device using MATLAB and Arduino.
- Fabricated and tested multiple prototypes, reducing size by 40% after user feedback.
- Integrated electronic components and validated system functionality through prototype testing.
- Presented at the Ohio State University Engineering Design Competition, where it ranked top 20% of 200+ teams.

Arduino Musical Instrument Controller

March 2026-May 2026

- Designed and built a sound system that allows the user to play 4 instruments on an IR remote.
- Integrated LCD screen, IR receiver, MP3 player module, and speakers to create an interactive instrument.
- Developed C++ code to process remote inputs, audio files, and customize the instrument. Millis function used to eliminate playback latency and improve real time responsiveness.
- Performed circuit analysis, component testing, and debugging to ensure correct use.

3D Printed F1 Toy Car

February 2024-May 2024

- Designed and built a battery powered toy F1 car using CAD models (Fusion 360) and 3D printing.
- Integrated electrical circuits, motors, 9V battery, and fan propulsion into the design using soldering.
- Developed an initial cardboard prototype and improved the design through iterative changes after testing.

SKILLS

Technical Skills: CAD (Onshape, Fusion 360), HVAC Installation Equipment, Power Tools, Soldering, Brazing, Bending, Pressing, Manual Lathe, Milling Machine, Reaming, Threading, Facing, 3D Printing, MATLAB, C++, QuickBooks, Excel.

EXTRACURRICULARS

American Society of Mechanical Engineers

Columbus, OH

Member

August 2025-Present

- Participated in networking events and opportunities with engineering professionals.